

STAT 405 Project Proposal

Hierarchical Bayesian Forecasting of U.S. State Unemployment Rates

Calvin Du (Student ID: 16507378)

March 25, 2026

1. Team

Calvin Du (Student ID: 16507378)

2. Code Repository

<https://github.com/calvingdu/stat405>

3. Scientific Questions

As a STAT + ECON major, I am particularly interested in labour markets, especially given the disruptions of the past several years including COVID-19, the rise of AI, and recent political changes. These have all materially affected employment patterns. Unemployment is probably the most consequential and closely-watched labour market indicator signalling a good or bad economic environment. State-level unemployment rates in the U.S. have huge variation mainly driven by local industry composition, geography, and sensitivity to national macroeconomic conditions (eg, California is more independent than other states). At the same time, federal monetary policy and typical economic growth cycles will always affect all the states regardless of how independent they are. This creates a need to balance treating each state independently and pooling all states together.

Primary question:

Can I produce more accurate unemployment forecasts for each state by partially pooling information across the states, borrowing strength from the national trend rather than purely fitting each state by themselves?

Secondary question:

Does including covariates like the federal monetary policy (like federal interest rates) and national GDP growth improve forecasts beyond an autoregressive model?

Both questions are naturally explored using Bayesian hierarchical models, where the degree of pooling is itself inferred from the data.

4. Datasets

4.1 Primary Dataset: BLS State Unemployment

The U.S. Bureau of Labor Statistics publishes monthly state-level unemployment rates through its Local Area Unemployment Statistics (LAUS) program. I've chosen to use this the "Employment status of the civilian noninstitutional population, seasonally adjusted" series because it is the most widely cited and it's very clean and well-maintained. With it, I have a long time series of unemployment rates for all 50 states.

Coverage: I'll be using all 50 U.S. states, from January 2000 to December 2025, monthly frequency (seasonally adjusted). This yields a 50×311 data matrix (we are missing 2025-10)

URL: <https://www.bls.gov/lau/rdscnp16.htm>

Dataframe structure (head):

FIPS	State	Year	Month	Labor force	Employed	Unemployed	Rate (%)
01	Alabama	1976	01	1,486,509	1,387,606	98,903	6.7
02	Alaska	1976	01	160,709	149,226	11,483	7.1
04	Arizona	1976	01	963,980	865,261	98,719	10.2
05	Arkansas	1976	01	889,912	825,233	64,679	7.3
06	California	1976	01	9,770,553	8,873,133	897,420	9.2
...							

4.2 Secondary Dataset: FRED Macroeconomic Covariates

In order to answer the secondary question, I will also pull in national macroeconomic covariates from the Federal Reserve Economic Data

For background, FRED is a widely used database of economic time series maintained by the Federal Reserve Bank of St. Louis. It's extremely useful for economic research and forecasting, and is used a lot by economists and data scientists.

The data I'll be using are:

- **Federal funds effective rate (FEDFUNDS) - Monthly:** This is the interest rate at which depository institutions trade federal funds with each other. It is a key tool of U.S. monetary policy that has a strong influence on economic activity and unemployment.
- **Real Gross Domestic Product (A191RL1Q225SBEA) - Quarterly:** This is the real gross domestic product, which is a measure of overall economic output. The growth rate of real GDP is closely related to unemployment.

Both the series are downloaded as plain CSV files directly from FRED (<https://fred.stlouisfed.org>) and cover January 2000 to December 2025.

observation_date	FEDFUNDS	observation_date	A191RL1Q225SBEA
2000-01-01	5.45	2000-01-01	1.5
2000-02-01	5.73	2000-04-01	7.5
...		...	
2025-12-01	3.72	2025-10-01	0.7

4.3 Backup Dataset: OECD Country Unemployment

As a backup, the Organisation for Economic Co-operation and Development (OECD) publishes unemployment data across various countries. An alternative to only focusing on U.S states is to apply the same hierarchical modelling approach to a broader set of regions across multiple countries.

The URL for this data is:

https://sdmx.oecd.org/public/rest/data/OECD.SDD.TPS,DSD_LFS@DF_IALFS_UNE_M,1.0/AUS+AUT+BEL+CAN+CZE+DNK+FIN+FRA+DEU+GRC+HUN+IRL+ITA+JPN+KOR+NLD+NZL+NOR+POL+PRT+SVK+SVN+ESP+SWE+CHE+TUR+GBR+USA..PT_LF_SUB..Z.Y..T.Y_GE15..M?startPeriod=2000-01&dimensionAtObservation=AllDimensions&format=csvfile

5. Methodology

5.1 Model 1: Independent AR(1) (Naive)

Each state is modelled independently with no pooling. For state s and time t :

$$\begin{aligned}\alpha_s &\sim \mathcal{N}(5, 3) && \text{(baseline level of unemployment is around 5\%)} \\ \beta_s &\sim \mathcal{N}(0.5, 0.3) && \text{(baseline autoregressive coefficient - moderate persistence)} \\ \sigma &\sim \text{Exp}(1) && \text{(shared SD across states)} \\ y_{s,t} \mid \alpha_s, \beta_s, \sigma &\sim \mathcal{N}(\alpha_s + \beta_s y_{s,t-1}, \sigma)\end{aligned}$$

This serves as the naive baseline. Since states are fit independently, estimates for small or volatile states will be noisy, with no regularisation from the national trend or neighbouring states. This model does allow states whose unemployment rates genuinely differ from the national picture to have their own parameters without being pulled toward the national mean.

5.2 Model 2: Hierarchical AR(1) with Covariates

State parameters are drawn from national priors, enabling partial pooling. Macroeconomic covariates enter the likelihood as shared predictors:

$$\begin{aligned}\mu_\alpha &\sim \mathcal{N}(5, 3) && \text{(national mean intercept)} \\ \sigma_\alpha &\sim \text{Exp}(1/2) && \text{(spread of intercepts across states)} \\ \mu_\beta &\sim \mathcal{N}(0.5, 0.3) && \text{(national mean persistence coefficient)} \\ \sigma_\beta &\sim \text{Exp}(1/0.3) && \text{(spread of persistence coefficients across states)} \\ \gamma &\sim \mathcal{N}(0, 1) && \text{(national economic covariate coefficients)} \\ \sigma &\sim \text{Exp}(1) && \text{(shared SD across states)} \\ \alpha_s \mid \mu_\alpha, \sigma_\alpha &\sim \mathcal{N}(\mu_\alpha, \sigma_\alpha) && \text{(state intercept, partially pooled)} \\ \beta_s \mid \mu_\beta, \sigma_\beta &\sim \mathcal{N}(\mu_\beta, \sigma_\beta) && \text{(state persistence coefficient, partially pooled)} \\ y_{s,t} \mid \alpha_s, \beta_s, \gamma, \sigma &\sim \mathcal{N}(\alpha_s + \beta_s y_{s,t-1} + \gamma^\top \mathbf{x}_t, \sigma)\end{aligned}$$

where $\mathbf{x}_t = (\text{FEDFUNDS}_t, \text{GDP_growth}_t)^\top$ contains two national covariates, with `GDP_growth` interpolated from quarterly to monthly. Unlike Model 1, α_s and β_s are tied together through shared hyperparameters $(\mu_\alpha, \sigma_\alpha, \mu_\beta, \sigma_\beta)$ which represent the national distribution of state-level parameters. This allows for shrinkage: states with more noise will be pulled toward the national mean, while states with strong signals can deviate more. The key variable is σ_β where a small value forces all states toward the same estimated national level coefficient μ_β , while a large value allows states to behave independently.

5.3 Posterior Inference

- **Method 1: SMC via simPPLe:**

SMC will be a good fit for the independent model where parameters are uncorrelated across states, and it will be interesting to see how it performs on the hierarchical model where parameters are correlated and more complex. However, it may struggle with the large number of parameters and the hierarchical structure, leading to slower convergence and less accurate estimates.

- **Method 2: MCMC/HMC via Stan**

MCMC is more flexible for complex hierarchical models and can handle the correlated parameters better. Also it will be faster to handle so many parameters and more efficient for the hierarchical model.

The comparison will examine:

- Posterior mean and credible interval and agreement between methods
- Computational cost including runtime and effective sample size (ESS) per second
- MCMC diagnostics such as $\hat{R}$, ESS

5.4 Model Comparison and Evaluation

The data are split like any timeseries: January 2000 to December 2023 will be set as training data, and January 2024 to December 2025 (24 months) will be the test set. Evaluation metrics:

- RMSE of the posterior predictive mean per state
- Shrinkage plot: posterior mean $\hat{\beta}_s$ (Model 2) vs. MLE from independent fits (Model 1)

5.5 Diagnostics and Validation

- $\hat{R}$ for all parameters
- Trace plots for hyperparameters $(\mu_\beta, \sigma_\beta)$ and selected state-level parameters
- Non-centred parameterisation if divergences arise in Stan

6. Potential Model Misspecifications

- **COVID-19:** unemployment spiked to $\sim 15\%$ nationally during 2020 and is an extreme outlier. A sensitivity analysis excluding 2020 will be reported.
- **Structural breaks:** the relationship between unemployment and covariates may change over time, especially with recent economic disruptions. A time-varying coefficient model could be explored as an extension.